

Matthew Glasser

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EDUCATION

Georgia Institute of Technology (GT), Atlanta, GA

May 2029

Bachelor of Science in Mechanical Engineering

Courses taken include: Physics I & II, Statics, Differential & Integral Calculus, Engineering Graphics, Linear Algebra

John Jay Senior High School, Hopewell Junction, NY

June 2026

ROBOTICS & ENGINEERING PROJECTS

FIRST Tech Challenge Team Jaybots #16700, *President*

Sept 2022 – June 2026

- Served as President, CAD Leader, & Manufacturer
- Designed and manufactured 50%+ of the robot's mass in custom parts, using CNC machining, waterjet cutting, and manual mill/lathe on steel and aluminum
- Applied design-for-manufacturability principles throughout, tailoring part geometry to available fabrication methods and tolerances
- Ran preliminary and critical design reviews (PDR/CDR), using cost-benefit analysis to prioritize design iterations and set goals between prototypes
- Created detailed manufacturing drawings with GD&T standards for CNC and manual-machined components
- Managed Bills of Materials across multiple robot iterations, tracking parts and budget
- Qualified for the 2025 World Championship (top 250 of 7000+ teams worldwide) and 2026 Canada Cup of Robotics
- Named FIRST Tech Challenge Dean's List Semifinalist (2025)
- Led 30+ outreach events and mentored two junior FTC/FLL teams
- Raised over \$120K in 4 years

RoboJackets: RoboWrestling, *Mechanical Team Member*

August 2026 – Present

- Designing and building an autonomous competitive sumo robot, including motor control and sensor integration
- Applying CAD and manufacturing skills (SolidWorks, CNC/manual machining, 3D printing) to iterate on a robot design
- Work closely with electrical and software team members to integrate sensors and autonomous motion algorithms

Embedded Robotics Projects, *Independent*

Ongoing

- Built motor controller and sensor integration projects using ESP32 microcontrollers, programmed in C++
- Designed and built an automatic EV charging cable retractor, driving a stepper motor and spool based on garage-door detection (TOF sensor) and vehicle API data
- Developed an autonomous maze-navigating robot using MicroPython for Science Olympiad
- Programmed competition robot behavior in Kotlin for FTC incorporating 24 sensors, including cameras for Computer Vision

LEADERSHIP

Eagle Scout & Senior Patrol Leader, Scouting America (Boy Scouts) Troop 182

March 2019 – June 2026

- 300+ community service hours
- Helped grow troop from 5 boy scouts to almost 20 coed scouts over 3 years
- Completed National Youth Leadership Training (NYLT), served on staff for 3 years

WORK EXPERIENCE

Ember Learning & DeAP Learning Labs, *Lead Web Developer*

Jan 2024 – Present

- Manage a team of 10+ developers, assigning tasks and reviewing code
- Develop User Interface and backend integrations between UI, databases, and AI models
- 200K users, \$120K+ revenue

PROJECTS

Etsy Store: 3D Mania Prints, *Founder & Owner*

2023 – Present

- Created & maintain shop page advertising and selling custom & personalized 3D printed designs (3dmaniaprints.etsy.com)
- Manage orders and customer conversations, coordinating with customers to design custom commissions and unique items
- 5.0-star rating

SKILLS

CAD: SolidWorks, OnShape, Fusion360, Blender

Manufacturing: 3D Printing (FDM & SLA), CNC Machining (Router, Waterjet, Mill), Manual Machining (Mill, Lathe)

Programming: Python, C++, Kotlin, MATLAB, MicroPython, Java, JavaScript